Computer Organization and Architecture

Assignment-1

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01. What is Raspberry Pi?

Ans.

- Raspberry Pi is a single board computer which contain all the basis components merged into one – processor , memory , input-output cip , graphic into a single circuit board.
- It was invented by Raspberry pi foundation in order to make people learn fundamentals of computer easily.

02. AI Summit 10 key Takeways.

Ans. AI summit was held at Bharat Mandapam in New Delhi from 16-21 February 2026, attended by industry leaders , policymakers , clinicians, researchers and technology executives.

Takeaways-

- **AI-Impact :** Instead of focusing on making AI has a powerful technology , the focus must be on creating it as efficient problem – solver in real-world . AI must be beneficial in solving problems of various industries like healthcare , education , public services etc.
- **AI for All –** Countries and communities should have affordable access to AI technologies and it should not be restricted to a certain level which includes – technology companies, wealthy countries , highly technical people etc.
  The summit focused on reducing the AI divide between communities and countries.
- **The three Sutras: People, Planet, Progress**- The summit emphasized that AI development should be accessible to all , benefit the society , protect the environment and contribute to sustainability.

- **Human Capital –** Countries need to invest in making AI as a fundamental skill through AI education , AI literacy , vocational training , etc. Every person should have basic level of knowledge of how AI works and how to use it effectively.
- **Multilingual AI –** AI systems should be effectively used by everyone . The summit focused on including multilingual and contextual evaluations to work across different languages and cultures.
- **Trustworthy and Responsible AI –** The summit emphasised on making AI safe , secure , reliable , transparent and accountable to achieve widespread adoption with trust.
- **Goverance Frameworks -** Appropriate governance frameworks must be created to protect people but should not restrict innovation.
- **Sustainable AI –** AI systems significantly require natural resources like electricity , computing resources , cooling and data centre infrastructure . Therefore , summit called for energy-efficient AI systems.
- **AI for Science and Social Development** – AI should be used in accelerating scientific research and social development .
- **From Global AI Competition to Global AI Cooperation-**AI development requires international collaboration. Countries, researchers, companies, and institutions should share knowledge, best practices, resources, and successful AI solutions so that innovations can be adapted and scaled across different regions and communities.

03. What is Yotta?

Ans.

- Yotta Data Services operates hyperscale data-center infrastructure in india. These are large scale physical infrastructure facilities that provide the computing foundation for cloud, AI, storage, networking and digital services.
- Yotta data centers majorly operate in Navi Mumbai, Greater Noida and GIFT City (Gujarat International Finance Tec-City).
- They sell their computing services to technology companies that need powerful, secure and scalable computing without having to build all that infrastructure themselves.

04. What is NPTEL?

Ans. NPTEL (National Programmeis an online education platfrom created by IITs and IISc, with funding from the Ministry of Education, Government of India. It provides university-level courses, especially covering engineering and computer-science subjects.

05. How to decide a programming language to design an application and what are parameters?

Ans. Selection of programming language depends on:

What type of application you want to build?

What are its requirements?

What are the frameworks required and resources available ?

Parameters:

Type of Application

Different languages are used to build different type of appliactions . For example:

Web-Application – Javascript , TypeScript , Java , PHP

AI/ML Application – Python

System Software – C/C++

Performance Requirement-

Selection of the language what the application requires in terms of performance .

Example – For high speed and low-level control , C may be preffered.

Libraries and frameworks-

For a requirement functionality , the language should have suitable libraries and frameworks. Example – in the field of AI and machine learning , python is used as it contains libraries like numpy , pandas etc.

06. RAM Size , Word Size , speed and operationg system of our PC.

Ans.

- **Computer:** 13th Gen Intel Core i5-1334U
- **Operating System**: Microsoft Windows 11
- **RAM Size**: 16.0 GB
- **Speed:** 3200 MT/s (MT = Mega Transfers)
- **Word Size** (The amount of data that a CPU can process in a single operation)
  64-bit (x64-based)

07. Why GPU is faster than CPU?

Ans. GPU is faster than CPU due to its architecture . A GPU has many more smaller processing units and can divide work into thousands of smaller parallel operations.

GPU can perform huge number of similar mathematical operations simultaneously.

08. CPU v/s TPU v/s GPU

Ans.

Feature	CPU	GPU	TPU
Full Form	Central Processing Unit	Graphic Processing Unit	Tensor Processing Unit

Main purpose	General-Purpose computing – performs many different types of operations.	Performs Parallel computing and graphics - doing many operations at the same time	AI/ML & tensor operations- specifically designed to speed up machine-learning calculations.
Used for	Running Operating System, Web browsing, running applications, etc.	Gaming ,AI/ML Learning, Image processing, etc.	AI model training, Deep learning, Large AI models,etc.
Example	Intel Core i5	NVIDIA H100	Google TPU v5e

09. Last year Turing Award

Ans. The A.M. Turing Award is one of the most prestigious awards in computer science field. Turing Award was awarded to Charles H. Bennett and Gilles Brassad for establishing the foundations of quantum information science and transforming secure communication and computing.

10. Stampeding herd problem

Ans. Stampeding herd problem occurs when multiple computers work together and when multiple tasks are making progress during the same period of time . A number of requests overloads the system as many threads , processes or clients try to access the same resource after it becomes available.

11. Who is the founder of intel , AMD?

Ans. Intel was founded by Robert Noyce and Gordon Moore.

AMD (Advanced Micro Devices) was founded by Jerry Sanders.

12. Which CPU organisation is our Laptop?

Ans.

- **CPU:** 13th Gen Intel Core i5-1334U
- **CPU Organisation:** General Register Organization

Modern Intel x86-64 processors use a register-based architecture with multiple general-purpose registers.